3D ToF Scanning System

Technical Datasheet and Application Note

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1. Device Overview

1.1 Features

- 3D spatial scanning using Time-of-Flight sensing
- 32-point angular resolution per scan
- System operates at a 36 MHz clock
- UART communication at 115200 baud
- I2C interface with VL53L1X sensor
- MATLAB-based 3D visualization
- User push-button (PJ0, PJ1)
- Estimated cost is approximately $50–$70 CAD, based on the microcontroller, ToF sensor, stepper motor, and driver components
- Onboard LED status indicators
- 3.3 V logic supply, 5 V motor supply
- Microcontroller uses on chip flash memory (1 MB) and SRAM (256 KB)

1.2 General Description

The 3D Time-of-Flight (ToF) Scanning System is an embedded system designed to measure distances and reconstruct a three dimensional representation of an indoor environment. The system uses a VL53L1X ToF sensor with a TI MSP432E401Y board to perform distance measurements at angular positions. A stepper motor is used to rotate the sensor, recording measurements over a full 360 degree range. The system collects data at 32 discrete angular points per scan, providing an angular resolution of approximately 11.25 degrees. Distance measurements are recieved through I2C communication between the sensor and the microcontroller, eliminating the need for an external ADC. The ToF sensor measures reflected light pulses and converts it into a corresponding distance value. Measured data is transmitted to a host computer using UART communication at a baud rate of 115200 bps. MATLAB is used to process the incoming data and convert it from polar to Cartesian coordinates for 3D visualization. Multiple scans can be performed to create layered representations of the environment. User interaction is achieved through onboard push buttons that control scan initiation and processing, while onboard LEDs provide overall status indication. The system operates using 3.3 V for logic and sensing components, and 5 V for the stepper motor, connected to a ULN2003 driver.

1.3 Block Diagram

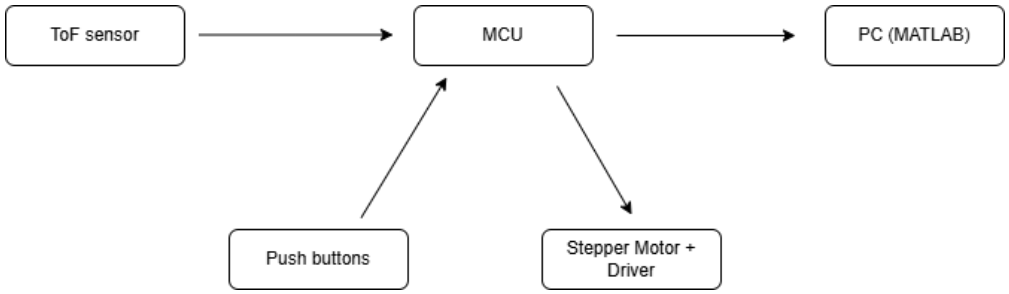

Figure 1: System Block Diagram

2. Device Characteristics

Parameter	Value
Supply Voltage (Logic)	3.3 V
Motor Supply Voltage	5 V
Clock Speed	36 MHz
UART Baud Rate	115200 bps
Sensor Range	Up to 4000 mm
Angular Resolution	32 points
Communication	I2C, UART
Scan Layers	User-defined
Operating Mode	Manual trigger (push buttons PJ0, PJ1)
Motor Output	Port L
I2C Communication	SDA (PB3), SCL (PB2)

Table 1: Device Characteristics

3. Detailed Description

3.1 Distance Measurement

Distance measurement is performed using a Time-of-Flight (ToF) sensor, which calculates the distance to an object by emitting a light pulse and measuring the time required for the reflected signal to return to the sensor. This method allows for accurate and reliable distance measurement.

$$d = \frac{c \cdot t}{2} \tag{1}$$

where c is the speed of light and t is the measured round trip time of the emitted signal. The division by 2 accounts for the departure and return travel of the light pulse.

The ToF sensor performs this calculation internally and outputs the measured distance in millimeters and does not require an ADC. The microcontroller interfaces with the sensor using the I2C communication protocol. It initiates a ranging sequence, waits for the sensor to indicate that data is ready, and then reads the measured distance from the sensor's internal registers. A complete scan consists of 32 individual measurements, each corresponding to a unique angular position.

3.1.1 I2C Communication

The I2C protocol operates using a master slave organization in which the microcontroller acts as the master device and the ToF sensor acts as the slave. Values are communicated through two lines, the clock line (SCL), and the data line (SDA). Once the measurement process is started, the system waits for data readiness, once ready, reads distances and records the values.

3.1.2 Stepper Motor Control

A stepper motor is used to rotate the ToF sensor, enabling distance measurements over a range of angles. The motor is controlled through a ULN2003 driveer, which allows the microcontroller to energize the motor coils in clockwise sequence. The motor completes one full 360 degrees rotation in 32 discrete steps, resulting in an angular resolution of 11.25 degrees per step. At each step, the motor pauses to allow the sensor to take a distance measurement. When 32 steps have been completed, the motor rotates counter-clockwise to untwist any wires and reset the system.

3.1.3 Data Acquisition

Distance measurements are transmitted over UART, with each value corresponding to a specific angular position in a scan. A complete dataset contains 32 measurements per scan cycle. Multiple scan cycles can be layered to build a more comprehensive representation of the chosen environment.

3.2 Visualization

Once the distance data is collected, it is transmitted from the microcontroller to a host computer using UART serial communication. The communication is configured at a baud rate of 115200 bits per second. The host computer, running the MATLAB script, receives the data and processes the measurements. The polar measurements are then converted into Cartesian coordinates as described above.This generates a 3D visualization of the scanned environment. MATLAB functions such as `scatter3` are used to plot the individual data points in 3D space, while `plot3` is used to connect points within each scan to form continuous rings. Multiple scan layers are combined to produce a complete 3D representation of the environment.

3.2.1 Data Processing and Conversion

The raw data recieved from the ToF sensor consists of distance measurements d in millimeters and their corresponding angular positions θ, determined by the stepper motor position. This data is initially collected by the microcontroller and transmitted to a host computer using UART communication. The conversion from polar coordinates (d, θ) to Cartesian coordinates (x, y, z) is performed on the computer using the MATLAB script. For each measurement, the angular position is calculated as:

$$\theta = \frac{2\pi}{32}(n-1) \tag{2}$$

where n is the measurement index and 32 represents the total number of steps per full rotation. The corresponding Cartesian coordinates are computed using:

$$y = d\cos(\theta), \quad z = d\sin(\theta) \tag{3}$$

To extend the 2D scan into a 3D representation, a displacement is introduced along the x-axis for each completed scan. The x-coordinate is defined as:

$$x = k \cdot \Delta x \tag{4}$$

where k is the scan index and Δx is the fixed displacement between consecutive scans.

3.2.2 Example Calculation

For a sample measurement where $d = 100$ mm and $\theta = 90°$:

$$y = 100 \cos(90°) = 0 \tag{5}$$

$$z = 100 \sin(90°) = 100 \tag{6}$$

If this measurement corresponds to the third scan ($k = 2$) with $\Delta x = 10$ mm:

$$x = 2 \cdot 10 = 20 \tag{7}$$

Thus, the final coordinate is $(x, y, z) = (20, 0, 100)$.

3.2.3 Displacement Implementation

Displacement represents the change in position between consecutive scans, allowing multiple 2D scans to be combined into a 3D representation. In this project, physical translation along the x-axis was not implemented. The displacement was simulated in MATLAB by assigning a constant offset Δx to each scan. This method creates a layered effect where each scan is positioned at a different x-location, allowing the reconstruction of a 3D environment from rotational measurements.

4. Application Note, Instructions, and Expected Output

4.1 Application Overview

This system is designed to scan an indoor environment using a Time-of-Flight sensor mounted on a stepper motor and generate a three-dimensional representation of the scanned area. The ToF sensor collects distance measurements at multiple angular positions, and the data is transmitted to a host computer using UART communication. MATLAB is then used to process the data and reconstruct a 3D model of the environment. The system was tested by scanning a campus hallway to evaluate performance and accuracy.

4.2 System Setup Instructions

4.2.1 Hardware Setup

1. Connect the ToF sensor to the microcontroller using I2C communication:
   - SDA → PB3
   - SCL → PB2
   - VIN → 3.3V
   - GND → Ground
2. Connect the stepper motor to the ULN2003 motor driver.

3. Connect the ULN2003 inputs to the microcontroller GPIO pins.
4. Power the system:
   - Connect microcontroller via USB
   - Stepper motor using a 5V external supply

4.2.2 Software Setup

1. Open the microcontroller IDE and load the program.
2. Open MATLAB on the host computer.
3. Set COM port.
4. Run the MATLAB script to begin data acquisition and visualization.

4.3 Operating Instructions

1. Power on the system.
2. Press onboard switch PJ0 to initiate scanning.
3. The stepper motor rotates the ToF sensor through 32 angular positions.
4. At each position, a distance measurement is taken.
5. The collected data is transmitted to the PC via UART communication.
6. MATLAB receives the data
7. Press onboard switch PJ1 to add another scan layer
8. Press PJ0 again to end scan
9. MATLAB processes data and generates a 3D model

4.4 Coordinate System Definition

The coordinate system used for 3D reconstruction is defined as follows:

- x-axis: represents displacement between consecutive scans
- y-axis: represents horizontal distance from the sensor
- z-axis: represents vertical distance based on angular position

4.5 Expected Output

The expected output of the system is a 3D point cloud representing the scanned environment. Each scan produces a ring of points in the yz-plane. Multiple scans are combined along the x-axis to form a layered 3D configuration. The resulting model should resemble the general geometry of the scanned location, including walls and open space.

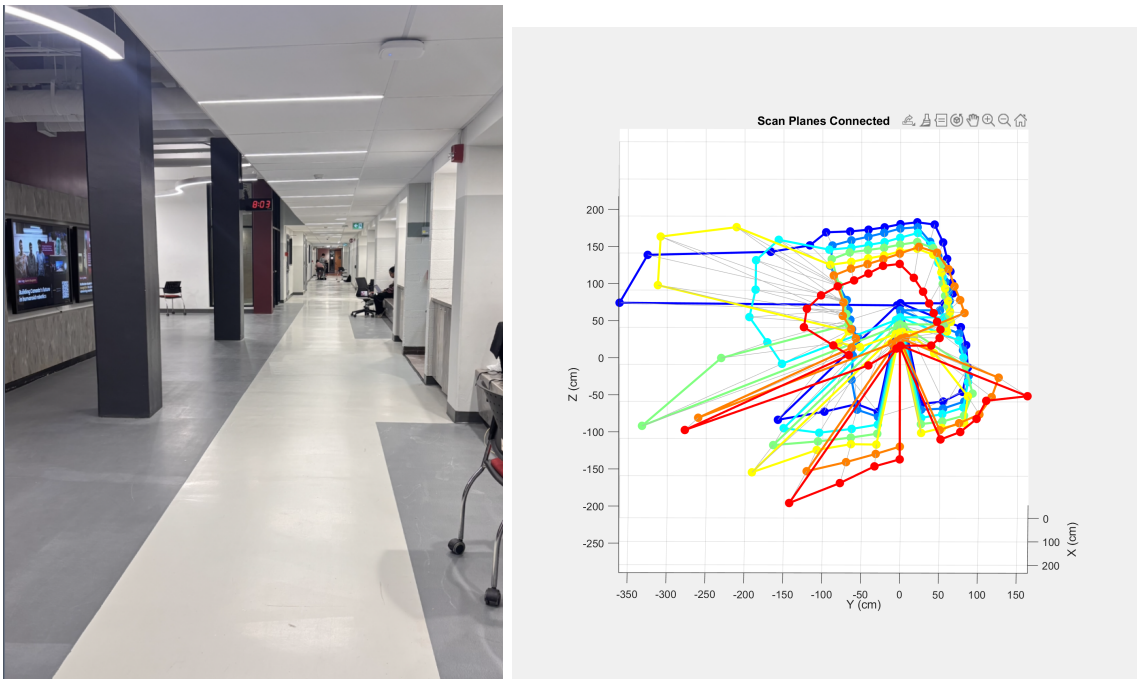

Figure 2: Comparison between the actual campus location and the reconstructed 3D model

4.6 Comparison with Real Environment

The reconstructed 3D model captures the overall structure of the environment, including the position of walls and other features. While some noise and distortion may be present due to sensor limitations and resolution, the general plot closely matches the real-world location.

4.7 User Notes

For optimal performance, the device should be placed on a stable surface to minimize vibration and measurement disturbance. Highly reflective or transparent surfaces may introduce measurement errors.

5. Limitations

- Maximum Communication Rate: The maximum reliable serial communication rate used between the microcontroller and the PC is 115200 baud. This rate was verified experimentally by transmitting data and observing the output in MATLAB. At higher baud rates, data corruption and instability were observed, while at 115200 baud, the data was consistently received without loss or errors.
- System Speed Limitations:
  - ToF Sensor Timing: The sensor requires a finite amount of time to perform each distance measurement, making it one of the primary bottlenecks.
  - Stepper Motor Movement: The motor must physically rotate and stabilize before each measurement, introducing delays between readings.

  - I2C Communication: Data transfer between the microcontroller and sensor adds additional latency.
  - UART Transmission: Serial communication to the PC further contributes to total system delay.

- Floating Point: MCU has limited floating point precision. Trig functions (sin/cos) introduce rounding error.
- Maximum range limited to 4000 mm due to ToF sensor
- Measurement noise at long distances and reflective surfaces
- No true linear translation (simulated X-axis)

These limitations were evaluated by measuring the total time required to complete a full scan and observing the effects of reducing delays between motor steps. It was found that reduced delays resulted in unstable measurements, confirming that sensor timing and motor stabilization are the main limiting factors. Overall, the ToF sensor measurement time and stepper motor response are the primary constraints on system performance.

6. Circuit Schematic

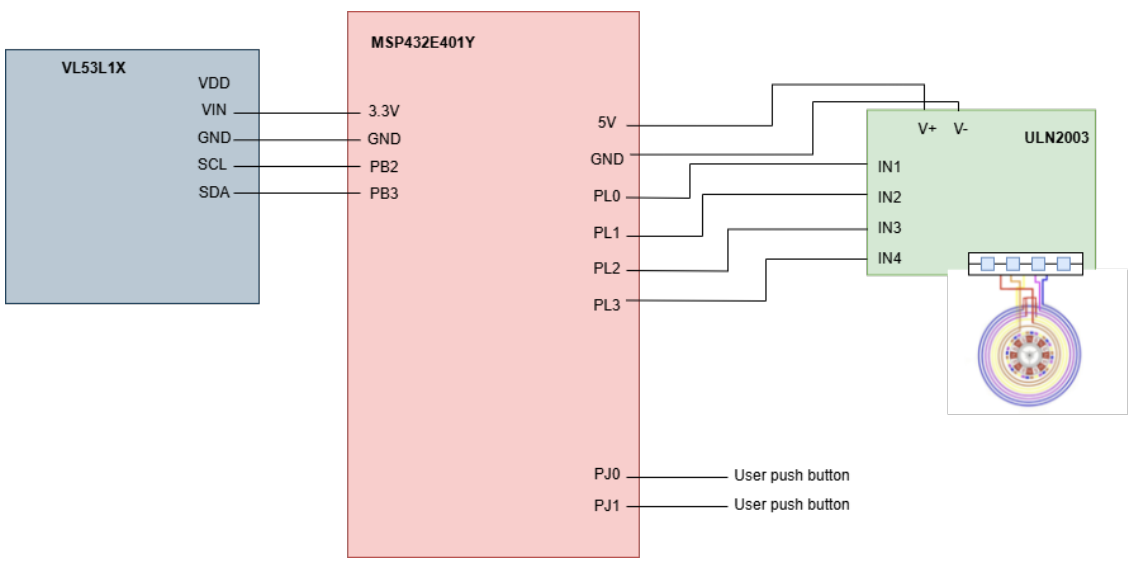

Figure 3: Circuit Schematic

7. Programming Flowcharts

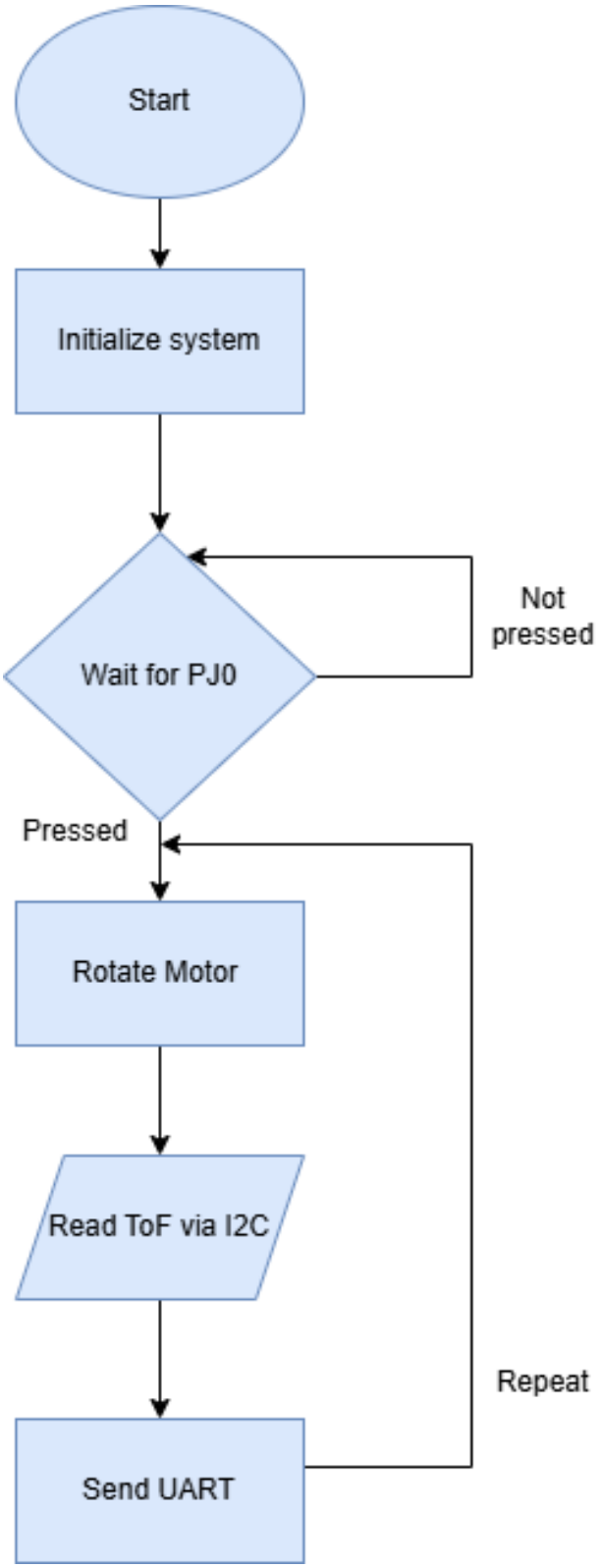

Figure 4: System Flowchart

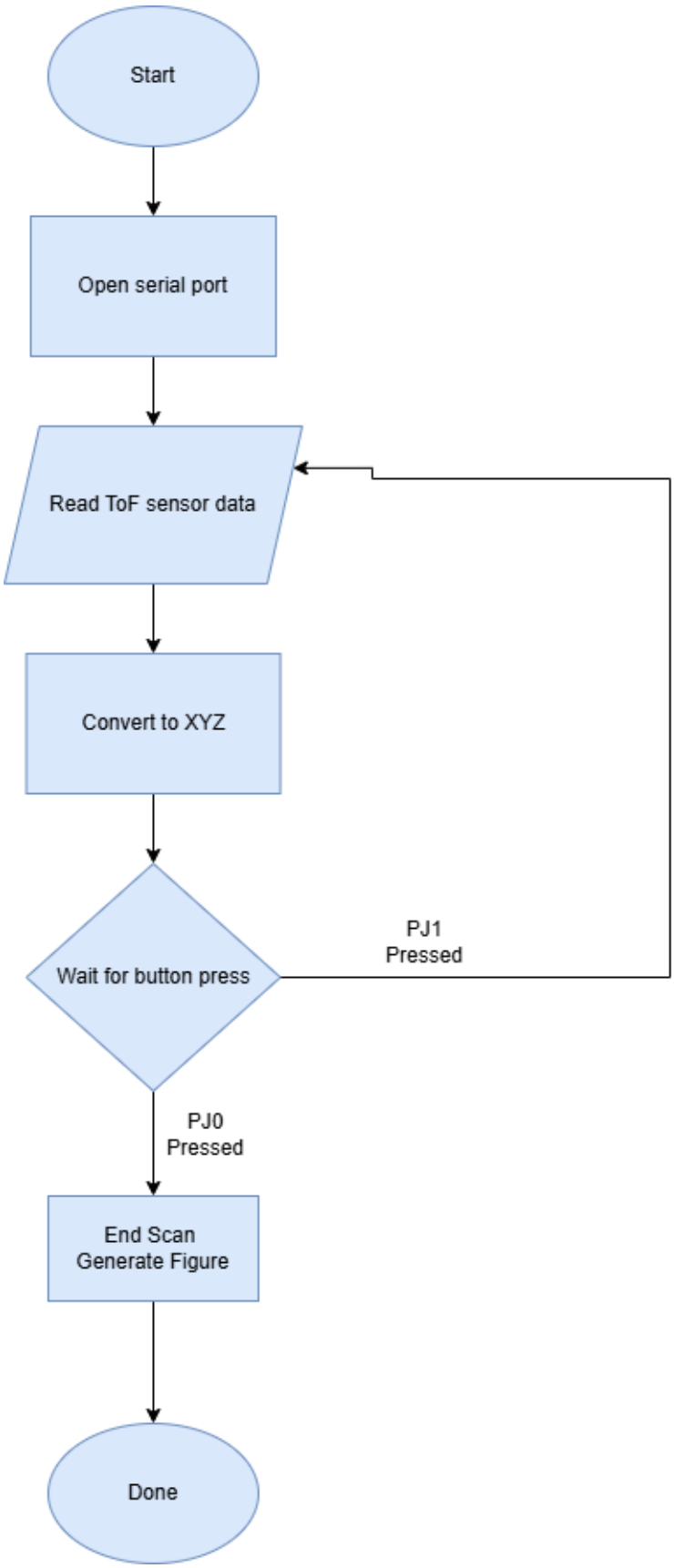

Figure 5: MATLAB Flowchart